ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** an acid	Acid.
2	**(a)** glow dimly	4.5 V is less than the rated 6 V.
3	**(d)** magnesium	Basic oxide, so metal.
4	**(a)** a reversible chemical change	Hydrated and anhydrous forms.
5	**(b)** 40 mL	Proportional.
6	**(a)** test with a tester; add salt to water to make it conduct	Dissolved salts supply ions.
7	**(c)** magnesium	Magnesium is highly reactive.
8	**(b)** formation of a new substance	New substance.
9	**(a)** acidic	Acid rain.
10	**(d)** a conductor	Current passes through it.
11	**(d)** zinc	Acid reacts.
12	**(c)** store in dry place or oil them	Keeps moisture away.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Turmeric turns red (reddish brown) with bases, and soap solution is a base, so R explains A.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Current flows only in a complete (closed) circuit, so R explains why the bulb glows.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Malleability is the property of being beaten into thin sheets, so R explains A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Lemon juice is acidic (1); It neutralises the base so turmeric turns back to yellow (1).	1+1
17	Tungsten has a very high melting point (1); Copper would melt before glowing (1).	1+1
18	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Zinc sulphate and hydrogen (1).	1+1
19	Iron combines with oxygen and water (1); Rust has extra mass (1).	1+1
20	Use litmus or turmeric (1); Different colours appear (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Ratio 10:20 (1); 60 is three times (1); 30 mL acid (1).	1+1+1
22	6 x 1.5 = 9 V (1); Much higher than 3 V (1); Bulb may fuse (1).	1+1+1
23	Metal (1); Magnesium (1); Basic oxide (1).	1+1+1
24	A: no rust (no moisture) (1); B: no rust (no air) (1); C: rusts (air and water) (1).	1+1+1
	OR: Some are more sensitive (1); Colour changes differ (1); Choose according to need (1).	1+1+1
25	They need a large current (1); Thin wiring may overheat (1); Risk of fire (1).	1+1+1
	OR: $MgO + H_2O \longrightarrow Mg(OH)_2$ (1); $SO_2 + H_2O \longrightarrow H_2SO_3$ (1); Basic and acidic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Hydrochloric acid (1). (ii) It is a base (an antacid) (1). (iii) Neutralisation (1). (iv) Lemon juice is acidic (it contains citric acid) (1), so it would add more acid to the stomach and make the burning worse instead of neutralising the extra acid (1).	1+1+1+2

27	Colour and lustre (1); Conduction test (1); Magnet for iron (1); Burning gives acidic or basic oxide (1); Reaction with acid (1).	1+1+1+1+1
	OR: Set-ups: dry, no air, tap water, salt water (1); Same nails and period (1); Observe daily (1); Compare rust amounts (1); Salt water most, oiled none (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** copper sulphate	Copper deposits.
2	**(b)** 30 mL	Doubling.
3	**(b)** tap water has dissolved salts	Dissolved ions conduct.
4	**(b)** Fe	Iron.
5	**(a)** lime water test	Test gas.
6	**(a)** it is a base used to relieve acidity	Carbonates react with acids.
7	**(b)** an insulator	Current does not pass.
8	**(a)** copper	Good conductor.
9	**(d)** apply oil and keep dry	Oiling.
10	**(b)** turns almost neutral	They neutralise.
11	**(a)** series	Single path.
12	**(c)** zinc	Zinc.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Only the state changes from solid to liquid, so R explains why it is physical.
14	**(d)** A is false but R is true.	Acids turn blue litmus red and do not change red litmus; so A is false, but R is true.
15	**(c)** A is true but R is false.	A fuse is in series, so A is true; but fuse wire has a LOW melting point and melts when current is too large, so R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Metals are good conductors (1); They carry current to the appliance (1).	1+1
17	They dissolve in rain to form acids (1); Such as SO_2 giving sulphurous acid (1).	1+1
18	Burning forms ash and gases (1); Tearing only changes size and shape (1).	1+1
19	Neither acid nor base (1); The effects of acid and base cancel out (1).	1+1
20	Prevent touching live parts (1); Avoid shocks (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Salt and sand do not react (1); Iron and sulphur react (1); New compound formed (1).	1+1+1
22	$H_2SO_4 + 2\,KOH \longrightarrow K_2SO_4 + 2\,H_2O$ (1); Acid and base (1); Salt and water (1).	1+1+1
23	Switch off the supply (1); Dry hands (1); Use proper bulb (1).	1+1+1
24	They are brittle or soft (1); Hammers need hard metal (1); Iron or steel used (1).	1+1+1
	OR: Burning: chemical (1); Light, sound, heat (1); Smoke and gases (1).	1+1+1
25	Add slaked lime (1); Add wood ash (1); Add powdered limestone (1).	1+1+1
	OR: Thin wire heats more (1); Melts first (1); Protects the circuit (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $1.5 + 1.5 = 3$ V (1). (ii) The switch (1). (iii) The filament (1). (iv) The filament may be broken (fused) or there may be a loose connection (1); in both cases the circuit is not complete, so no current flows through the filament (1).	1+1+1+2
27	Burn Mg ribbon (1); Dissolve ash in water (1); Burn sulphur and dissolve gas (1); Test both with litmus (1); Compare (1).	1+1+1+1+1

	OR: Burn a candle in a closed jar (1); It goes out (1); Add lime water (1); It turns milky (1); Conclude (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** blue to green	$CuSO_4$ is blue.
2	**(d)** unburnt carbon, a chemical change	New substance.
3	**(a)** a carbonate and an acid	CO2 is released.
4	**(a)** both conduct, copper better	Brightness shows current.
5	**(b)** Q	Reacts fastest.
6	**(c)** to prevent rusting and reduce friction	Both reasons.
7	**(a)** red, purple, blue	Acid to base.
8	**(c)** a single path for the current	The loop is broken.
9	**(b)** less reactive than copper	Copper displaces it.
10	**(c)** the wax changes into new substances such as carbon dioxide and water vapour	Mass is conserved.
11	**(a)** the acid was stronger or more concentrated	Excess acid.
12	**(c)** the nail is coated with paint or rust	Surface layer blocks contact.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but being a form of carbon does not explain why graphite conducts.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but being a metal does not explain the need for oxygen and water.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the release of heat does not explain why salt and water form.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	They conduct heat away from the hand (1); Heat flows quickly (1).	1+1
17	Takes a long time (1); Small changes each day (1).	1+1
18	A mucus layer protects it (1); The acid is dilute (1).	1+1
19	Nichrome gets very hot when current passes (1); It does not melt or oxidise easily at high temperatures (1).	1+1
20	Gold is very unreactive (1); Silver tarnishes with sulphur (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Colour (1); Density (copper is heavier) (1); Conductivity (1).	1+1+1
22	Chemical (1); New substance slaked lime (1); Heat released (1).	1+1+1
23	Neutralisation releases heat (1); It is exothermic (1); Salt and water form (1).	1+1+1
24	Iron would melt or burn (1); Tungsten has a high melting point (1); Bulb has inert gas (1).	1+1+1
	OR: Soft (1); Reactive with air (1); Surface tarnishes (1).	1+1+1
25	Burning candle (1); Melting is physical (1); Burning vapour is chemical (1).	1+1+1
	OR: Add equal juice (1); Compare the red colour depth (1); More intense means more acidic (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Metals are good conductors of heat (1). (ii) Plastic is a poor conductor of heat, so the handle does not become hot (1). (iii) Malleability (1). (iv) Ductility (1); copper is also a good conductor of electricity (1).	1+1+1+2

27	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); $Mg + CuSO_4 \longrightarrow MgSO_4 + Cu$ (1); Mg is reactive (1); Products observed (1).	1+1+1+1+1
	OR: LPG burns: chemical (1); Milk warms: physical (1); Milk boils: physical (1); Milk burns at bottom: chemical (1); Black deposit is carbon (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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